MILESTONE 1 SUBMISSION - CMPG325-2026-065

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Client: North-West Electrical Engineering Consultants (Klerksdorp)
Client ID: CLI-065
Assigned Addressing Block: 192.168.34.0/24

PART 1: CLIENT REQUIREMENTS

1.1 Who Is the Client?

I am designing this network for North-West Electrical Engineering Consultants. They are an engineering company based in Klerksdorp that provides electrical consulting services to different clients. Like many businesses today, they need a reliable network to keep their daily operations running smoothly.

From what I understand about their work, they need to:

. Access engineering design software and tools

. Share large technical documents and project files

. Communicate with clients and other stakeholders

. Run presentations in their boardroom

. Handle administrative tasks like payroll and HR records

The company needs a stable and secure network that works well now but can also grow if they hire more staff or expand their offices.

1.2 What Does the Client Need?

After going through the project brief carefully, I identified these main requirements:

Network Connectivity:

. All staff members need to be able to connect their computers and devices to the network

. Different departments should be able to share information when needed

. The boardroom must have both wired and wireless internet access for presentations

IP Address Planning:

. I must use the address block 192.168.34.0/24 that was assigned to me

. I need to divide this block into smaller subnets for different departments

. I must leave some addresses unused so the client can expand later

Boardroom Constraint:

. The boardroom needs dedicated Ethernet ports where people can plug in their laptops or presentation PCs

. There must also be Wi-Fi available for visitors who come with tablets or phones

. Both wired and wireless need to work at the same time

Security:

. I must configure ACLs (Access Control Lists) to filter traffic between different parts of the network

. This means I decide what traffic is allowed and what is blocked

. For example, I might block the boardroom from accessing the HR department's computers

Remote Management (CR9):

. One IT administrator needs to be able to manage the network from outside the office

. They must use SSH which is secure, not Telnet which is not secure

. Only this one person should have this remote access

Implementation:

. I need to build a working network in Cisco Packet Tracer

. Everything must be tested and proven to work

. I must document all my work on GitHub

1.3 Explaining the Boardroom Constraint

The project says the boardroom needs both wired and wireless presentation ports. This is what I think it means:

. Wired ports mean Ethernet wall jacks where someone can plug in a presentation PC or laptop for a stable connection

. Wireless access means a Wi-Fi access point in the boardroom so visitors can connect without cables

I decided to put all boardroom devices in their own VLAN (VLAN 30). This way:

Boardroom traffic stays separate from internal departments

It's more secure because visitors can't access sensitive company data

It's easier to manage and troubleshoot if something goes wrong

1.4 The ACL Challenge Explained Simply

My assigned networking challenge is ACLs (Access Control Lists). In simple terms, ACLs are like security guards at the door of each department. They decide:

. Who is allowed to enter

. Who is not allowed

. What type of traffic can pass through

I will configure these ACLs on the Layer 3 core switch. This means:

. Engineering staff can access the servers they need

. Boardroom users can only access the internet, not internal departments

. Management can access everything (they need to)

. The remote admin can only access management interfaces

Later when I test everything, I will verify that:

. Devices that should communicate can do so

. Devices that should be blocked are blocked

. The security rules work as expected

1.5 What About the Remote Admin? (CR9)

Change Request 9 says one off-site administrator needs secure access to manage network devices. Here's how I planned to handle this:

. I created a separate remote management VLAN (VLAN 99) just for this purpose

. The admin will use SSH which encrypts all traffic so nobody can snoop

. I will restrict SSH access to only the admin's IP address

. ACLs will make sure the admin can only access management interfaces, not user data

When I test this in Packet Tracer, I will show that:

. The remote admin can successfully connect using SSH

. The admin can manage switches and routers

. Unauthorized users cannot connect

2.1 Why I Chose This Design

When I started planning this network, I had to decide between two main approaches. After thinking about it and listening to what our lecturer said, I went with a switch-based design rather than a traditional router setup.

Why I chose switch-based design:

REASON	MY EXPLANATION
Speed	Layer 3 switches can route traffic really fast because they do it in hardware. A router would be slower because it processes routing in software.
Scalability	If the company grows, I can just add more switches. With a router-based design, it would be harder to expand.
Cost	One Layer 3 switches both switching and routing. This means the client doesn't need to buy two separate devices.
modern approach	Most companies today use Layer 3 switching. This is what the lecturer wants us to learn.

2.2 The Core of My Network

The most important device in my design is the Layer 3 core switch. This device does a lot of the heavy lifting:

What It Does	Why It Matters
Connects all access switches together	All departments can reach each other
Routes traffic between VLANs	Devices in different departments can communicate when they need to
Enforces ACL rules	This is where the security policies are applied
Hosts management interfaces	I can manage all network devices from one place

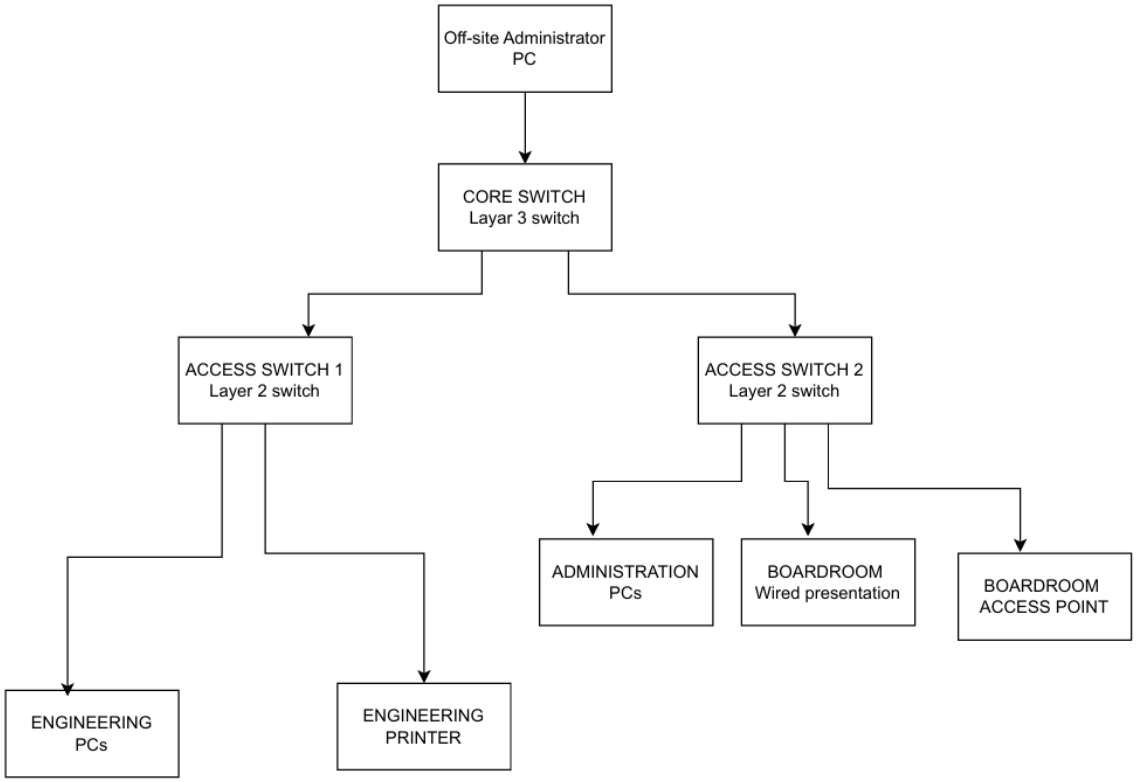
2.3 Access Switches and End Devices

I placed Layer 2 access switches in each department to connect the end devices:

Switch	Location	What it connects
SW-ENG	Engineering Department	20 engineering PCs
SW-HR	HR	15 ADMINISTRATIVE PCs
SW-BOARD	BOARDROOM	2 PRESENTATION PCs + WAP
SW-MGMT	MANAGEMENT OFFICE	3 MANAGEMENT PCs
SW-SERVER	SERVER ROOM	DHCP/FILE SERVER

PART 3: PHYSICAL LAYOUT

My Physical Topology Diagram



Device Placement Table

Room/Area	Device	Type	VLAN
Server Room	R1	Router (WAN only)	N/A
Server Room	SW-CORE	Layer 3 Switch	All VLANs
Server Room	Server-1	DHCP/File Server	VLAN 40
Engineering	SW-ENG	Layer 2 Switch	VLAN 10
Engineering	PC-ENG-01 to 20	Workstations	VLAN 10
HR/Admin	SW-HR	Layer 2 Switch	VLAN 20
HR/Admin	PC-HR-01 to 15	Workstations	VLAN 20
Boardroom	SW-BOARD	Layer 2 Switch	VLAN 30
Boardroom	PC-BOARD-1, 2	Presentation PCs	VLAN 30
Boardroom	WAP	Wireless Access Point	VLAN 30

Management	SW-MGMT	Layer 2 Switch	VLAN 50
Management	PC-MGMT-1 to 3	Management PCs	VLAN 50
Remote	Admin PC	Remote Administrator	VLAN 99

How things are connected

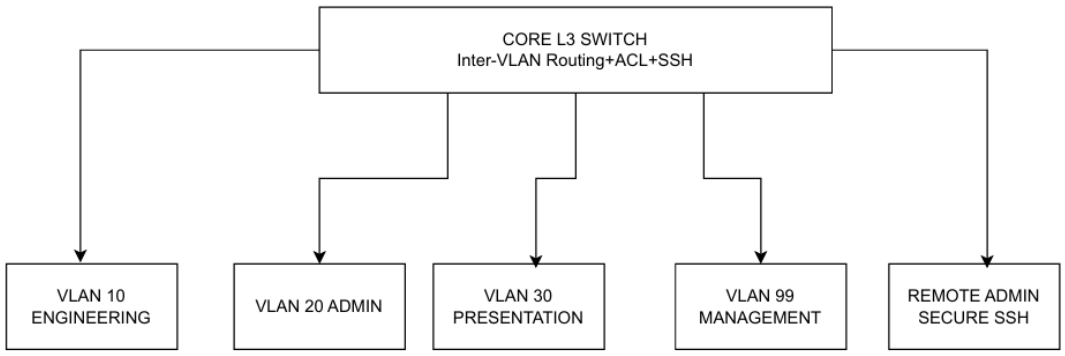
Connection	Cable Type
R1 (Gig0/1) to SW-CORE (Gig0/1)	Copper Straight-through
SW-CORE (Gig0/2) to SW-ENG (Gig0/1)	Copper Straight-through
SW-CORE (Gig0/3) to SW-HR (Gig0/1)	Copper Straight-through
SW-CORE (Gig0/4) to SW-BOARD (Gig0/1)	Copper Straight-through
SW-CORE (Gig0/5) to SW-MGMT (Gig0/1)	Copper Straight-through
SW-CORE (Gig0/6) to Server-1 (Fast0)	Copper Straight-through
SW-ENG (Fa0/1-20) to PCs (1-20)	Copper Straight-through
SW-HR (Fa0/1-15) to PCs (1-15)	Copper Straight-through
SW-BOARD (Fa0/1) to PC-BOARD-1	Copper Straight-through
SW-BOARD (Fa0/2) to PC-BOARD-2	Copper Straight-through
SW-BOARD (Fa0/3) to WAP	Copper Straight-through
WAP to Laptop-1, Laptop-2	Wireless (Wi-Fi)
SW-MGMT (Fa0/1-3) to PCs (1-3)	Copper Straight-through

PART 4: LOGICAL DESIGN (VLANs)

I divided the network into VLANs so I could separate different types of traffic. This makes the network more organized and secure.

VLAN ID	Name	What's In It	Subnet
10	Engineering	Engineering workstations	192.168.34.0/27
20	HR-Admin	HR and admin staff	192.168.34.32/27
30	Boardroom	Presentation devices	192.168.34.64/28
40	Servers	Server infrastructure	192.168.34.80/28
50	Management	Network device management	192.168.34.96/28
99	Native	Untagged traffic	N/A

My Logical Topology Diagram



My ACL plan

I will apply ACLs on the Layer 3 core switch using these interfaces:

Interface	Direction	What It Does
VLAN 10 SVI	Inbound	Blocks Engineering from accessing HR
VLAN 20 SVI	Inbound	Blocks HR from accessing Engineering
VLAN 30 SVI	Inbound	Only allows Boardroom to access internet
VLAN 40 SVI	Inbound	Only allows specific hosts to access servers
VLAN 50 SVI	Inbound	Management can access everything

PART 5: IP ADDRESSING PLAN

I started with the assigned block 192.168.34.0/24 and divided it into these subnets:

VLAN	Name	Network	Prefix	Usable Range	Broadcast
10	Engineering	192.168.34.0	/27	.1 - .30	.31
20	HR-Admin	192.168.34.32	/27	.33 - .62	.63
30	Boardroom	192.168.34.64	/28	.65 - .78	.79
40	Servers	192.168.34.80	/28	.81 - .94	.95
50	Management	192.168.34.96	/28	.97 - .110	.111

Why I Chose Each Subnet Size

VLAN	Prefix	Why I Chose It
Engineering	/27 (30 hosts)	There are 20 engineers now. This leaves room for 10 more if they hire more staff.

HR-Admin	/27 (30 hosts)	There are 15 admin staff now. This gives room for 15 more later.
Boardroom	/28 (14 hosts)	Only 4 devices now. 14 addresses are more than enough for presentations.
Servers	/28 (14 hosts)	Only 1 server now. Plenty of room for more servers.
Management	/28 (14 hosts)	Only 3 management devices now. Room for 11 more.

Default Gateways

VLAN	Network	Gateway	Device
10	192.168.34.0/27	192.168.34.1	SW-CORE
20	192.168.34.32/27	192.168.34.33	SW-CORE
30	192.168.34.64/28	192.168.34.65	SW-CORE
40	192.168.34.80/28	192.168.34.81	SW-CORE
50	192.168.34.96/28	192.168.34.97	SW-CORE

Device IP Addresses

Device	VLAN	IP Address	Gateway
Engineering PC 1	10	192.168.34.10	192.168.34.1
Engineering PC 2	10	192.168.34.11	192.168.34.1
Engineering PC 3-20	10	.12 - .29	192.168.34.1
HR PC 1	20	192.168.34.40	192.168.34.33
HR PC 2	20	192.168.34.41	192.168.34.33
HR PC 3-15	20	.42 - .54	192.168.34.33
Boardroom PC 1	30	192.168.34.70	192.168.34.65
Boardroom PC 2	30	192.168.34.71	192.168.34.65
WAP	30	192.168.34.73	192.168.34.65
Laptop 1	30	DHCP	192.168.34.65
Laptop 2	30	DHCP	192.168.34.65

Server-1	40	192.168.34.90	192.168.34.81
SW-CORE Management	50	192.168.34.100	192.168.34.97
SW-ENG Management	50	192.168.34.101	192.168.34.97
SW-HR Management	50	192.168.34.102	192.168.34.97
SW-BOARD Management	50	192.168.34.103	192.168.34.97
SW-MGMT Management	50	192.168.34.104	192.168.34.97
Management PC 1	50	192.168.34.110	192.168.34.97
Management PC 2	50	192.168.34.111	192.168.34.97
Management PC 3	50	192.168.34.112	192.168.34.97
Remote Admin	99	192.168.34.130	192.168.34.129

PART 6: WHY I MADE THESE CHOICES

I chose a Layer 3 core switch instead of using a router for inter-VLAN routing. Here's why:

Reason	My Explanation
Speed	Layer 3 switches route at hardware speed. A router would be a bottleneck because it processes everything in software.
Cost	One Layer 3 switch does both jobs. The client saves money by not buying a separate router.
Growth	If the company expands, I can easily add more switches. A router-based design would be harder to scale.
Modern	This is how real companies design their networks. It's what our lecturer wants us to learn.

Why VLANs?

Reason	Why It Matters
Less broadcast traffic	Each VLAN has its own broadcast domain, so traffic doesn't flood the whole network.
Better security	Departments are isolated from each other. HR can't accidentally see Engineering files.
Easier management	I can apply policies to each VLAN separately.
ACL implementation	VLAN boundaries let me apply traffic filters effectively.

How I Designed the Boardroom

The boardroom is special because of the wired and wireless requirements. Here's what I did:

Feature	What I Did
Separate VLAN	Boardroom devices are on VLAN 30, away from internal departments.
Wired ports	Two Ethernet ports for stable, high-speed connections.
Wireless	A Wi-Fi access point for laptops and mobile devices.
Security	ACLs will block boardroom access to internal departments.
DHCP	Boardroom devices get IP addresses automatically.

How I Handled Remote Management (CR9)

For Change Request 9, I designed this approach:

Requirement	My Solution
Off-site admin	Dedicated remote admin on VLAN 99.

Secure access	SSH encryption, not Telnet.
Device management	Management VLAN 50 for all device management.
Access control	ACLs restrict SSH to the admin's IP only.

My ACL Strategy

Rule	Source	Destination	What I'll Do
1	Engineering (VLAN 10)	HR (VLAN 20)	BLOCK - Engineering should not access HR
2	Engineering (VLAN 10)	Servers (VLAN 40)	ALLOW - Engineering needs server access
3	Boardroom (VLAN 30)	Internal networks	BLOCK - Visitors should not access internal data
4	Boardroom (VLAN 30)	Internet	ALLOW - Visitors need internet access
5	Management (VLAN 50)	All VLANs	ALLOW - Management needs full access
6	Remote Admin (VLAN 99)	VLAN 50	ALLOW SSH ONLY - Admin can manage devices